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Faculty of Engineering
Electronics and Communications Engineering Department

GRADUATION PROJECT THESIS — PART II

Machine Learning Techniques for PAPR Reduction in Multicarrier Communication Systems

Project Team

Nour-Eldin Mohamed Mostafa	Youssef Samir Sleem Ahmed
Muhammad Mahmoud Muhammad	Reem Nader Saeed
Basel Mohamed Ramadan	Ahmed Walid Hassan
Yosef Ahmed Mohamed	

Supervisor

Dr. Michael Ibrahim

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Declaration of Authorship

We, Nour-Eldin Mohamed Mostafa, Youssef Samir Sleem Ahmed, Muhammad Mahmoud Muhammad, Reem Nader Saeed, Basel Mohamed Ramadan, Ahmed Walid Hassan, Yosef Ahmed Mohamed, declare that this thesis titled “Machine Learning Techniques for PAPR Reduction in Multicarrier Communication Systems” and the work presented in it are our own. We confirm that this material, submitted for assessment on the programme leading to the award of Bachelor of Engineering, is entirely our own work. We have exercised reasonable care to ensure that the work is original and does not, to the best of our knowledge, breach any law of copyright; it has not been taken from the work of others except to the extent that such work has been cited and acknowledged within the text.

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Abstract

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CHAPTER 1

OFDM System Model and Classical PAPR Reduction Techniques

To establish the requisite mathematical framework for this thesis, this chapter provides a preliminary review of the Orthogonal Frequency-Division Multiplexing (OFDM) signal model and the Peak-to-Average Power Ratio (PAPR) problem. It also examines classical PAPR reduction techniques and their inherent limitations, thereby motivating the shift toward the machine learning approaches explored in subsequent chapters.

1.1 OFDM Signal Model

Orthogonal Frequency-Division Multiplexing divides a wideband channel into N narrowband, orthogonal subcarriers. Each subcarrier carries one modulated symbol from a constellation such as QPSK or M -QAM. The modulation and demodulation operations are efficiently implemented via the Inverse Fast Fourier Transform (IFFT) and Fast Fourier Transform (FFT), respectively.

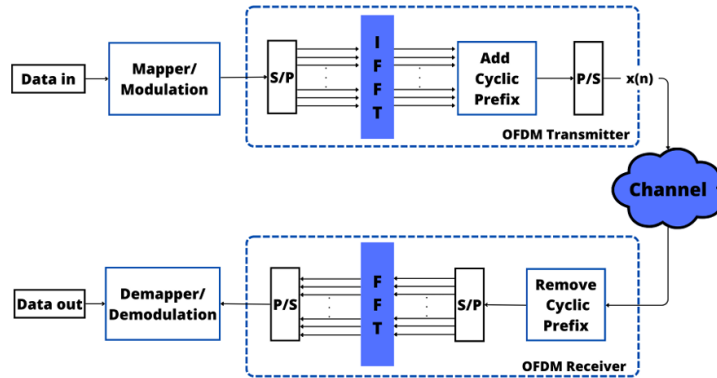


FIGURE 1.1: Block diagram of a baseband OFDM transmitter and receiver. The transmitter maps data onto N subcarriers via an N -point IFFT, adds a cyclic prefix (CP), and transmits the time-domain signal. The receiver removes the CP and applies an N -point FFT to recover the frequency-domain symbols [1].

1.1.1 Subcarrier Modulation via the IFFT

Let the frequency-domain data symbol vector after constellation mapping be

$$\mathbf{X} = [X[0], X[1], \dots, X[N-1]]^T,$$

where $X[k]$ is the complex data symbol mapped to the k -th subcarrier and N is the total number of subcarriers.

The time-domain OFDM signal is obtained by applying the N -point IFFT:

1.1

$$x[n] = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}, \quad n = 0, 1, \dots, N-1.$$

At the receiver, the original frequency-domain symbols are recovered by the complementary FFT operation:

1.2

$$X[k] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1.$$

The symmetric $1/\sqrt{N}$ normalization used in both of these transforms preserves signal energy between the frequency and time domains (Parseval's theorem).

1.1.2 Cyclic Prefix

Before transmission over the wireless channel, a **Cyclic Prefix (CP)** of length N_{cp} samples is prepended to each time-domain OFDM symbol by simply copying the final N_{cp} samples of the IFFT output and attaching them to the front. The CP serves two critical purposes:

- **ISI elimination:** provided N_{cp} exceeds the maximum channel delay spread, the CP completely absorbs all multipath echoes originating from the previously transmitted symbol.
- **Circular convolution:** the CP mathematically converts the linear convolution with the channel into a circular convolution, enabling simple single-tap equalization in the frequency domain via the FFT operation.

1.2 Peak-to-Average Power Ratio

1.2.1 Definition

The time-domain OFDM signal is the superposition of N independently modulated complex exponentials. When these subcarrier contributions align constructively at certain time instants, large amplitude peaks occur. The severity of this phenomenon is quantified by the **Peak-to-Average Power Ratio (PAPR)**:

1.3

$$\text{PAPR}(x) = \frac{\max_{0 \leq n \leq N-1} |x[n]|^2}{E[|x[n]|^2]}.$$

In this formulation, the numerator represents the peak instantaneous power and the denominator represents the average power of the OFDM symbol. The theoretical worst-case PAPR for N subcarriers is $10 \log_{10}(N)$ dB, which occurs when all subcarriers align perfectly in phase.

1.2.2 CCDF Performance Metric

Since PAPR varies from one OFDM symbol to the next, it is characterised statistically using the **Complementary Cumulative Distribution Function (CCDF)**:

$$\text{CCDF}_{\text{PAPR}} = \Pr(\text{PAPR} > \text{PAPR}_0),$$

where PAPR_0 is a given threshold. For N subcarriers with independent, identically distributed symbols and Nyquist-rate sampling, the CCDF can be approximated as:

1.4

$$\Pr(\text{PAPR} \geq \text{PAPR}_0) \approx 1 - (1 - e^{-\text{PAPR}_0})^N.$$

A PAPR reduction technique shifts the CCDF curve to the left, indicating that high-PAPR events become statistically less likely.

1.2.3 Power Amplifier Nonlinearity

High PAPR has direct consequences for the **High-Power Amplifier (HPA)** at the transmitter. Every power amplifier has a saturation region: when the input signal exceeds a certain amplitude, the output clips or compresses, introducing nonlinear

distortion. To avoid entering this nonlinear zone, the HPA must operate with sufficient **Input Back-Off (IBO)**:

$$\text{IBO (dB)} = 10 \log_{10} \left(\frac{P_{\text{sat}}}{P_{\text{avg}}} \right),$$

where P_{sat} is the amplifier's saturation power and P_{avg} is the average input signal power. To prevent severe nonlinear distortion, the required IBO must be tightly bounded by the signal's maximum PAPR.

1.3 Classical PAPR Reduction Techniques

Classical PAPR reduction techniques can be broadly classified into **signal distortion** methods, which introduce controlled modifications to the transmitted signal, and **signal scrambling** methods, which generate alternative signal representations without distortion.

1.3.1 Signal Distortion Techniques

Clipping and Filtering

In this straightforward approach, any time-domain signal sample that exceeds a predefined amplitude threshold is deliberately clipped. Because this nonlinear operation inevitably introduces out-of-band spectral regrowth, the clipped signal must subsequently be filtered to meet spectral mask requirements, though this introduces some in-band distortion.

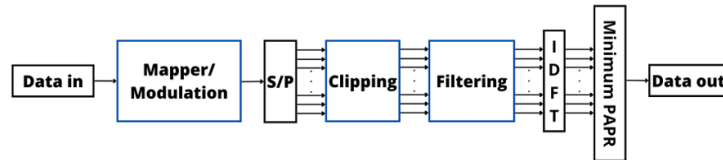


FIGURE 1.2: Block diagram of an OFDM transmitter with clipping and filtering. The signal is clipped in the time domain, then filtered in the frequency domain to remove out-of-band distortion [1].

While clipping and filtering is simple to implement and requires no side information, it introduces in-band distortion (increasing BER) and out-of-band radiation.

Tone Reservation (TR)

Tone Reservation sets aside a specific subset of subcarriers exclusively for generating peak-cancelling signals. This conceptually straightforward approach reduces PAPR without introducing any distortion to the actual data symbols, though it inevitably results in a slight reduction in overall spectral efficiency.

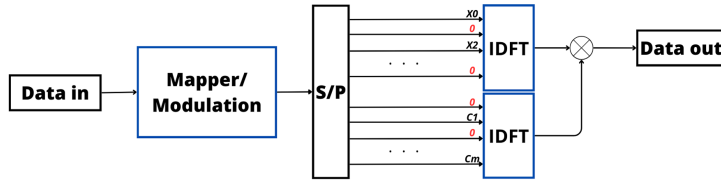


FIGURE 1.3: Block diagram of an OFDM transmitter with Tone Reservation (TR). A designated subset of subcarriers is reserved exclusively for generating peak-cancelling signals [1].

1.3.2 Signal Scrambling Methods

Signal scrambling methods alter the signal representation *without distortion* to find a low-PAPR version.

Partial Transmit Sequence (PTS)

The input data block is partitioned into disjoint sub-blocks, each transformed by an independent IDFT into the time domain. These sub-blocks are then individually weighted by different phase factors and summed together. The transmitter evaluates multiple phase combinations and transmits the one that yields the lowest overall PAPR.

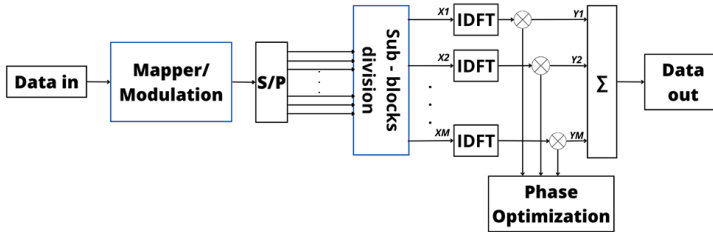


FIGURE 1.4: Block diagram of an OFDM transmitter with Partial Transmit Sequence (PTS). Data is split into M sub-blocks, each weighted by a phase factor after the IDFT. The combination with the lowest PAPR is transmitted [1].

Selected Mapping (SLM)

Selected Mapping generates multiple alternative representations of the same data block by multiplying the original frequency-domain symbols with several distinct, pseudo-random phase sequences. Each candidate sequence is transformed into the time domain via an independent IDFT, and the candidate exhibiting the absolute lowest PAPR is selected for transmission. To allow the receiver to correctly undo the phase rotation and recover the original data, the index of the chosen phase sequence must be transmitted alongside the signal as **side information (SI)**.

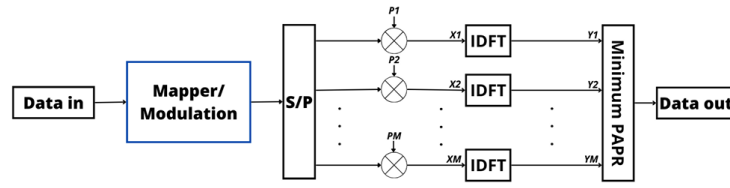


FIGURE 1.5: Block diagram of an OFDM transmitter with Selected Mapping (SLM). M different phase sequences are applied to the frequency-domain data before the IDFT. The candidate signal with the lowest PAPR is selected for transmission [1].

1.4 Limitations of Classical Approaches

Despite their ability to bound the peak power of OFDM signals, classical PAPR reduction techniques suffer from five fundamental limitations:

- **Distortion and spectral leakage:** Distortion-based methods, such as clipping and filtering, physically alter the time-domain waveform. This inevitably introduces in-band distortion that degrades the Bit Error Rate (BER) and causes out-of-band radiation that interferes with adjacent frequency channels.
- **Spectral and power inefficiency:** Methods like Tone Reservation (TR) avoid distortion by sacrificing a dedicated subset of subcarriers exclusively for peak-cancelling signals. This permanently reduces the system's spectral efficiency (data rate) and wastes transmission power on non-information-bearing signals.
- **Computational complexity:** Scrambling methods require multiple IFFT operations per OFDM symbol. For example, generating M candidate signals requires a per-symbol complexity of $\mathcal{O}(M \cdot N \log_2 N)$, which becomes prohibitive for large systems.
- **Signaling overhead:** Methods requiring side information (SI) must transmit $\lceil \log_2 M \rceil$ SI bits per symbol. These bits must be received error-free, necessitating additional channel coding that further reduces spectral efficiency.
- **Blind detection difficulty:** Eliminating SI overhead through blind detection at the receiver requires M full FFT operations per symbol, and classical decision metrics become highly unreliable at low Signal-to-Noise Ratios (SNR).

These limitations define the core design objectives for the machine learning-based approaches explored in subsequent chapters. By shifting from algorithmic search to data-driven optimization, deep learning models can achieve effective PAPR reduction while fundamentally overcoming the computational and signaling constraints of classical methods.

CHAPTER 2

PRNet: PAPR Reduction via Deep Autoencoder

The fundamental limitations of the classical PAPR reduction approaches identified in Chapter 1 motivate a paradigm shift in transceiver design. Rather than relying on rigid algorithmic searches or intentional signal distortion, Kim et al. [2] proposed the **PAPR Reduction Network (PRNet)**, the first deep learning-based system for joint PAPR and BER optimisation in OFDM. PRNet treats the transmitter–channel–receiver chain as a single deep autoencoder: the encoder learns a data-dependent constellation mapping that jointly minimises reconstruction error and PAPR, while the decoder learns the corresponding inverse mapping, fundamentally bypassing the computational and signaling bottlenecks of classical methods.

2.1 Autoencoder Framework

In a traditional communication system, the transmitter transforms raw data into a physical signal through a rigid sequence of fixed mathematical operations, which the receiver attempts to run in reverse after the signal is distorted by the wireless channel. This physical reality maps perfectly onto the architecture of a deep learning **autoencoder** [2]. In this framework, the neural network's *encoder* acts as the transmitter, the *decoder* acts as the receiver, and the mathematical layer connecting them acts as the noisy wireless channel. By placing a mathematical simulation of the channel directly between the encoder and decoder, the entire communication chain becomes a single, unified neural network.

The profound advantage of this unified architecture is that it enables **end-to-end joint training**. Rather than optimising the transmitter and receiver as isolated digital signal processing blocks, PRNet optimises them simultaneously. This allows the network to dynamically learn a custom, data-dependent constellation mapping at the encoder that directly minimises both the reconstruction error (BER) and the peak power (PAPR), while the decoder simultaneously learns the exact inverse mapping required to recover the data despite physical channel impairments [2].

To realise this, PRNet integrates two trainable Deep Neural Networks (DNNs) into the communication chain. At the transmitter, the encoder DNN is placed before the IFFT, replacing the standard constellation mapper. Instead of mapping each symbol to a fixed constellation point independently, the encoder employs a fully connected structure where every output subcarrier value is a function of the *entire* input block. By processing all subcarriers jointly, the encoder detects input patterns that produce high PAPR and compensates by adjusting all subcarrier amplitudes and phases to generate low-peak waveforms [2].

At the receiver, the decoder DNN is placed after the FFT to replace the standard constellation de-mapper. Mirroring the transmitter, it employs a fully connected structure to process the received OFDM block jointly. Due to joint training, the decoder's weights implicitly encode the encoder's mapping. Consequently, it reconstructs the original data from the equalised frequency-domain subcarriers without explicit side information, eliminating both the bandwidth overhead and severe failure modes associated with SI errors in classical methods [2].

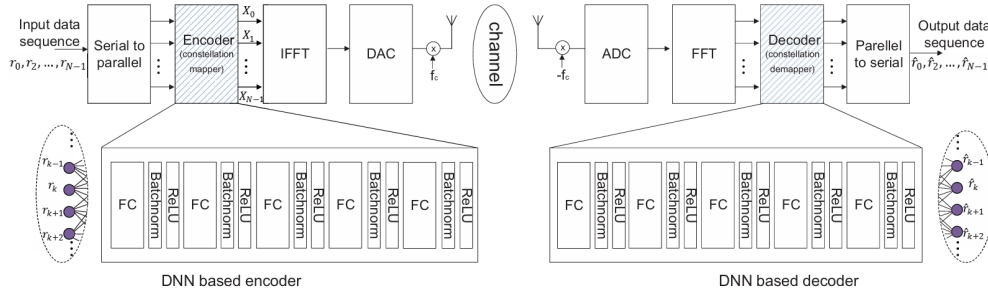


FIGURE 2.1: Structure of the PRNet system with DNN encoder and DNN decoder. Both consist of fully connected (FC) layers with Batch Normalization and ReLU activation. The encoder shapes the constellation before the IFFT to reduce PAPR; the decoder reconstructs the transmitted symbols after the channel and FFT [2].

2.1.1 Core Neural Network Components

Internally, both the encoder and decoder are constructed from identical repeated sub-blocks [2]. Each sub-block consists of three fundamental deep learning operations applied in sequence:

- **Fully Connected (FC) Layer:** Unlike standard OFDM where each subcarrier is mapped independently, the FC layers apply a linear transformation that combines the entire input vector. This dense connectivity enables the network to process all subcarriers jointly, recognising block-wide patterns that cause high PAPR and adjusting all subcarrier phases and amplitudes simultaneously.

- **Batch Normalization (BN):** Inserted after the linear transformations, BN standardises the intermediate constellation representations across each training batch. This stabilisation is critical for PRNet, as it prevents vanishing gradients and allows the network to converge on a highly complex, multidimensional constellation mapping without training collapse.
- **ReLU Activation:** The Rectified Linear Unit provides the essential nonlinearity. Without it, the entire multi-layer network would collapse into a single linear matrix projection, which is mathematically insufficient to discover the highly non-convex mappings required to jointly minimise PAPR and BER.

2.2 System Model

With the core neural network components established, we now formulate the PRNet transceiver as an end-to-end OFDM communication system [2]. The system is mathematically defined for N orthogonal subcarriers and an input sequence of M -bit symbols. For training and evaluation, we adopt the baseline configuration from Kim et al. [2], specifically $N = 64$ subcarriers and QPSK modulation ($M = 2$).

2.2.1 Input Representation

The original PRNet formulation [2] maps input sequences using 256-dimensional one-hot encoded vectors. This work replaces this categorical approach with a continuous real-valued in-phase and quadrature (I/Q) representation. Because standard deep neural networks operate exclusively in the real domain, they cannot process complex-valued OFDM symbols natively. To construct the necessary real-valued input, the 64 complex data symbols (each representing 2 bits) are explicitly decomposed into their in-phase (I) and quadrature (Q) components. The input sequence is thus defined as the 128-dimensional real-valued vector:

$$\mathbf{r} = [I_0, Q_0, I_1, Q_1, \dots, I_{N-1}, Q_{N-1}]^T \in \mathbb{R}^{2N}.$$

While mathematically equivalent to the one-hot formulation in achieving optimal BER and PAPR, the continuous I/Q parameterization is adopted to provide three strict architectural advantages. First, it reduces the input dimension from 256 to 128, strictly halving the parameter complexity of the initial projection matrix. Second, bypassing categorical encoding eliminates discrete mapping layers, allowing the network to optimize continuous physical coordinate geometry directly. Finally, it enables explicit per-symbol power normalization during the forward pass. This guarantees the strict physical constraint $\mathbb{E}[|X|^2] = 1$ for every individual transmitted waveform, overcoming the reliance on global batch-wide statistical approximations.

2.2.2 DNN Encoder

The transmitter is parameterized by the deep neural network $f(\cdot; \theta_f)$, where θ_f denotes the complete set of trainable weights and biases. The encoder follows a topology of $L_f = 5$ consecutive hidden layers, each with a width of 2048 nodes [2]. Crucially, its final output layer is a pure linear projection with no non-linear activation or batch normalisation [2]. This is physically necessary because the network must map symbols to unconstrained coordinates on the complex I/Q plane, which inherently requires the ability to output unbounded negative and positive spatial values.

The encoder maps the real-valued input vector $\mathbf{r}$ directly to a frequency-domain vector representing the generated complex constellation points:

$$\mathbf{X} = f(\mathbf{r}; \theta_f) = [X_0, X_1, \dots, X_{N-1}]^T \in \mathbb{C}^N.$$

2.2.3 OFDM Modulation

The generated frequency-domain symbols X_k are modulated onto the $N = 64$ orthogonal subcarriers via an Inverse Fast Fourier Transform (IFFT) to produce the discrete time-domain sequence $x[n]$.

The Peak-to-Average Power Ratio (PAPR) of this time-domain sequence is mathematically defined as the ratio of its maximum instantaneous power to its average power:

$$\text{PAPR}\{x[n]\} = \frac{\max_{0 \leq n \leq N-1} |x[n]|^2}{\mathbb{E}[|x[n]|^2]}.$$

Since standard Nyquist-rate sampling ($L = 1$) often fails to capture the true continuous-time peaks, accurate evaluation requires an oversampling factor of $L \geq 4$. In practice, this is executed by zero-padding the frequency-domain vector $\mathbf{X}$ to length NL before computing the IFFT. This step ensures the discrete digital PAPR reliably bounds the maximum power of the physical analog waveform.

2.2.4 Channel Model

Following the IFFT operation, the time-domain signal is transmitted through a wireless channel before arriving at the receiver. As in the original PRNet formulation [2], the physical transmission is abstracted using a holistic operator $\mathbb{H}$ that models the combined effects of multipath fading and thermal noise.

2.2.5 OFDM Demodulation and Equalization

At the receiver, OFDM demodulation is executed by applying an $N = 64$ -point Fast Fourier Transform (FFT) to convert the received time-domain sequence $y[n]$ back into the frequency-domain symbols Y_k .

Following the FFT, the transmission is modeled as a Rayleigh flat-fading environment. Under this model, the received signal on the k -th subcarrier is mathematically defined as $Y_k = H_k X_k + E_k$, where H_k represents the complex channel fading coefficient and E_k represents the additive white Gaussian noise (AWGN).

To compensate for the fading effects prior to neural network decoding, Zero-Forcing (ZF) equalization is applied. Assuming perfect Channel State Information (CSI)—meaning the receiver perfectly knows H_k —the equalized symbols are obtained by dividing the received signal by the channel coefficient:

$$\hat{Y}_k = \frac{Y_k}{H_k} = X_k + \frac{E_k}{H_k}.$$

2.2.6 DNN Decoder

The receiver is parameterized by the deep neural network $g(\cdot; \theta_g)$, where θ_g denotes the complete set of trainable weights and biases. The decoder shares a symmetric topology with the encoder, comprising $L_g = 5$ consecutive hidden layers, each with a width of 2048 nodes [2]. Similarly, its final output layer is a pure linear projection with no non-linear activation. This is required to allow the network to reconstruct unconstrained I/Q coordinate values from the noisy equalized signal prior to hard-decision demodulation.

The decoder maps the equalized frequency-domain vector $\hat{\mathbf{Y}}$ back to the 128-dimensional real-valued reconstructed sequence $\hat{\mathbf{r}}$. A hard-decision demodulator is subsequently applied to $\hat{\mathbf{r}}$ to recover the original transmitted bits.

Consequently, the complete end-to-end reconstructed symbol vector at the receiver can be written as a composite function of the entire transceiver pipeline:

$$\hat{\mathbf{r}} = g \circ \text{FFT} \circ \mathbb{H} \circ \text{IFFT} \circ f(\mathbf{r}),$$

where $\mathbb{H}$ abstractly denotes the physical effects of the wireless channel. Since every block in this composite chain is mathematically differentiable, gradients from the training loss function flow backward through the entire system during joint optimization of the encoder and decoder parameters.

2.3 Training Methodology

2.3.1 Loss Function

PRNet optimises two competing objectives simultaneously: reconstruction accuracy (low BER) and peak suppression (low PAPR). While the original PRNet formulation utilizes Cross-Entropy loss for its categorical one-hot encoded inputs [2], our continuous I/Q parameterization requires a regression-based loss. Thus, the reconstruction loss is formulated as the Mean Squared Error (MSE) between the original continuous input vector and the reconstructed output vector in the $2N$ -dimensional real space:

$$\mathcal{L}_1(\mathbf{r}, \hat{\mathbf{r}}) = \frac{1}{2N} \|\mathbf{r} - g(\text{FFT} \circ \mathbb{H} \circ \text{IFFT}(f(\mathbf{r}; \theta_f)); \theta_g)\|_2^2 = \frac{1}{2N} \sum_{i=0}^{2N-1} (r_i - \hat{r}_i)^2.$$

The PAPR penalty directly measures the peak-to-average power ratio of the time-domain waveform $x[n]$ [2]. To explain how this penalty actually trains the neural network, we mathematically trace $x[n]$ back to its source: the encoder. By expanding $x[n]$ into $\text{IFFT}(f(\mathbf{r}; \theta_f))$, the formula explicitly shows how the PAPR loss back-propagates through the IFFT block to optimize the encoder's internal weights θ_f :

$$\mathcal{L}_2(\mathbf{r}) = \text{PAPR}\{\text{IFFT}(f(\mathbf{r}; \theta_f))\} = \frac{\max_{0 \leq n \leq NL-1} |x[n]|^2}{\frac{1}{NL} \sum_{n=0}^{NL-1} |x[n]|^2}.$$

Finally, the complete joint loss function is a weighted combination of both the reconstruction loss and the PAPR penalty [2]:

$$\mathcal{L}(\mathbf{r}, \hat{\mathbf{r}}) = \mathcal{L}_1(\mathbf{r}, \hat{\mathbf{r}}) + \lambda \mathcal{L}_2(\mathbf{r}),$$

where $\lambda > 0$ is a weighting hyperparameter that governs the fundamental physical trade-off between reconstruction accuracy and PAPR reduction. A higher λ forces the network to aggressively reduce PAPR, but doing so requires distorting the transmitted constellation. This distortion shrinks the minimum distance between adjacent constellation points, which inherently degrades the decoder's ability to safely recover symbols under noise, leading to a higher BER. Therefore, the chosen value of λ dictates the balance of this trade-off.

2.3.2 Two-Stage Training Procedure

To force the decoder to learn robust decision boundaries, Kim et al. employ denoising autoencoder training, where artificial noise at a corruption level η (the noise-to-signal ratio) is injected during the forward pass (the simulated transmission of the signal through the physical channel before reaching the decoder) [2]:

$$\eta = \frac{\mathbb{E}[|\epsilon|^2]}{\mathbb{E}[|\mathbf{r}|^2]}.$$

Selecting η poses a critical trade-off: insufficient noise yields overly tight decision boundaries that fail in real channels, while excessive noise prevents distinguishing adjacent constellation points. To resolve this and evaluate optimal parameters, Kim et al. established a two-stage training procedure [2]:

1. **Stage 1 — Corruption level selection:** Multiple models are trained using solely the reconstruction loss $\mathcal{L}_1$ (setting $\lambda = 0$) across a sweeping range of η values. The models are evaluated across a full SNR sweep to empirically determine the optimal noise injection level η^* , as shown below.

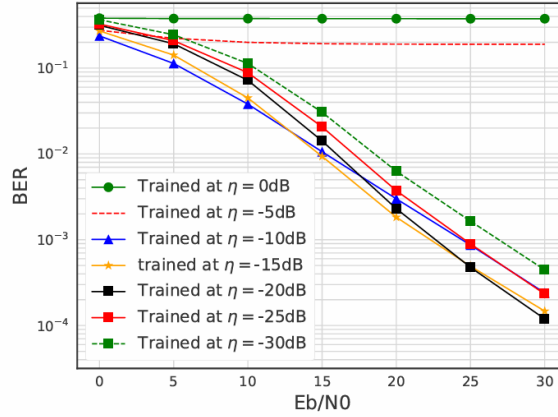


FIGURE 2.2: BER of PRNet trained at different corruption levels η . The model trained at $\eta = -15$ dB consistently yields the lowest BER [2].

The authors' results physically validate the noise trade-off: models trained at excessive corruption levels (e.g., $\eta = 0$ dB and $\eta = -5$ dB) fail to learn meaningful boundaries entirely, resulting in flat BER curves. Conversely, models trained at very low corruption levels (e.g., $\eta = -30$ dB) overfit to the noise-less scenario, yielding sub-optimal BER performance under real channel conditions. As the corruption level diverges in either direction from the optimal point, the overall BER steadily increases. Based on these empirical findings, Kim et al. conclude that $\eta^* = -15$ dB strikes the perfect balance [2].

2. **Stage 2 — Joint training:** The noise injection is fixed at $\eta^* = -15$ dB, and models are trained using the full joint loss $\mathcal{L} = \mathcal{L}_1 + \lambda \mathcal{L}_2$ to investigate how the penalty weight λ dictates the trade-off between PAPR reduction and BER degradation, as illustrated below.

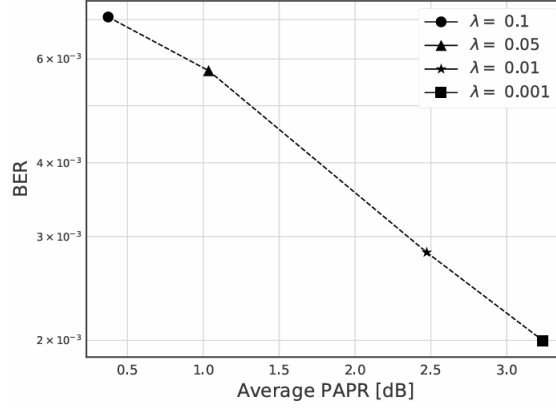


FIGURE 2.3: BER vs. average PAPR by varying λ at SNR = 20 dB. Larger λ reduces PAPR but increases BER [2].

As established by Kim et al., as λ increases, PAPR decreases monotonically while the BER increases, confirming the fundamental physical tension between constellation shaping and minimum distance. Heavy penalties (e.g., $\lambda = 0.1$) aggressively squash the waveform but severely compromise the error rate, whereas minimal penalties (e.g., $\lambda = 0.001$) prioritize a low BER at the expense of higher PAPR. Following this optimization sweep, the authors identified $\lambda = 0.01$ as the ideal practical operating point—striking a balance that achieves meaningful PAPR reduction without severely degrading the BER baseline [2].

2.4 Simulation

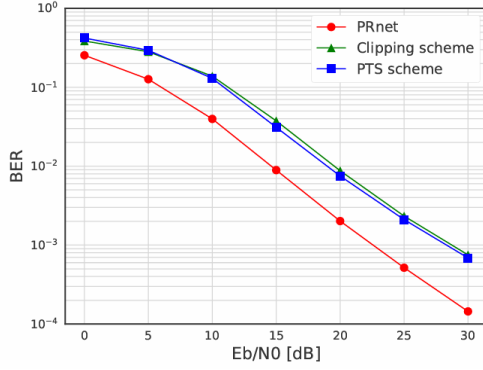
2.4.1 Simulation Parameters

TABLE 2.1: PRNet simulation parameters [2].

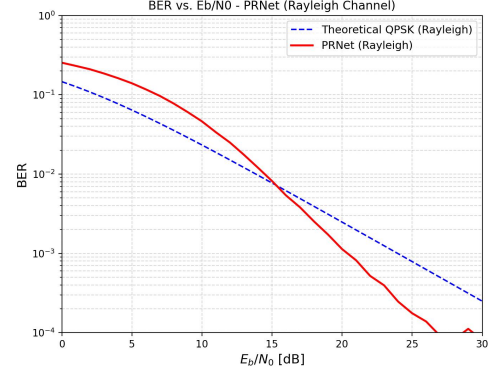
Parameter	Value
Number of subcarriers (N)	64
Modulation	4-QAM/QPSK
Network input representation	$2N = 128$ real entries (I/Q components)
Hidden processing blocks per DNN	5
Hidden nodes per layer	2048
Activation function	ReLU (hidden), Linear (output)
Batch normalisation	Yes, after every FC layer
Optimiser	Adam
Learning rate	10^{-4}
Batch size	400
Training steps	500,000
Optimal corruption level (η^*)	-15 dB
PAPR weight (λ)	0.01
Evaluation size	50,000 random OFDM sequences

Note that while the original paper denotes ReLU for all layers, our implementation utilizes continuous I/Q representation and MSE loss rather than one-hot classification. Therefore, a Linear activation is physically required at the final layer to allow the network outputs to span the full complex plane (both positive and negative values).

2.4.2 BER Performance



(A) Original paper's evaluation [2]



(B) Reproduced via our implementation

FIGURE 2.4: BER vs. SNR performance of PRNet. Both the (a) original paper and (b) our reproduction demonstrate consistent BER performance. The results confirm that PRNet effectively learns an implicit joint source-channel coding over the simulated channel, allowing it to outperform the theoretical baseline.

As observed in Figure 2.4, the BER performance of our reproduced model is highly consistent with the results reported by the original authors. PRNet acts as a highly efficient deterministic function requiring zero iterative searching or side information during inference. Furthermore, the network achieves a BER lower than the theoretical QPSK baseline because the encoder shapes the constellation in a way that the jointly trained decoder can exploit. This effectively learns an implicit coding gain under the simulated channel, validating the robustness of the deep learning approach [2].

2.4.3 PAPR Reduction

The CCDF evaluation demonstrates a substantial reduction in the PAPR of the PRNet-shaped symbols. The authors report that the probability of the PAPR exceeding 3 dB is less than 10^{-3} [2]. However, a mathematical critique of the authors' open-source code (https://github.com/gomman/OFDM_PAPR_reduction) reveals a metric discrepancy: the CCDF is plotted using an amplitude-based proxy rather than the standard power-based definition. Specifically, their implementation computes the metric as:

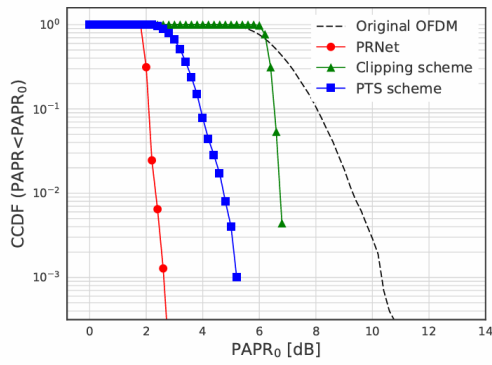
$$\text{PAPR}_{\text{proxy}} [\text{dB}] = 10 \log_{10} \left(\max_n |x_{\text{norm}}[n]| \right),$$

where $x_{\text{norm}}[n]$ represents the time-domain symbol normalized to unit average power. In contrast, the standard physical PAPR (in dB) is a true power ratio, defined mathematically as:

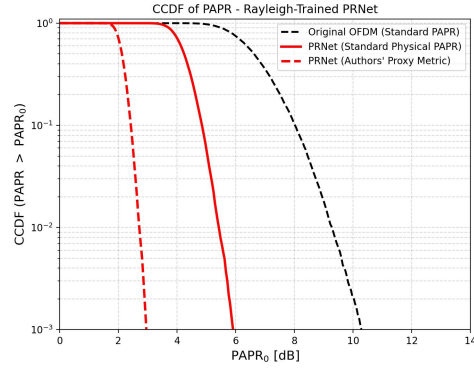
$$\text{PAPR}_{\text{standard}} [\text{dB}] = 10 \log_{10} \left(\max_n |x_{\text{norm}}[n]|^2 \right) = 20 \log_{10} \left(\max_n |x_{\text{norm}}[n]| \right).$$

Consequently, the proxy metric is mathematically exactly half of the standard physical PAPR in decibels ($\text{PAPR}_{\text{standard}} [\text{dB}] = 2 \cdot \text{PAPR}_{\text{proxy}} [\text{dB}]$).

As shown in Figure 2.5, evaluating the model using the authors’ proxy metric perfectly reproduces the 3 dB curve reported in the original paper, confirming the consistency of our reproduced architecture. However, evaluating the trained model under the standard physical metric yields a realistic PAPR of approximately 6 dB at a CCDF of 10^{-3} . This still represents a substantial 4–5 dB reduction from the original OFDM baseline (typically 10–11 dB), validating PRNet’s exceptional PAPR reduction capabilities despite the metric discrepancy.



(A) Original paper’s evaluation [2]



(B) Reproduced via our implementation

FIGURE 2.5: CCDF of PAPR performance for PRNet. Both the (a) original paper and (b) our reproduction demonstrate a consistent 3 dB curve when using the authors’ proxy metric. However, our reproduction validates that the standard physical metric yields a realistic 6 dB PAPR, which still significantly outperforms the original OFDM baseline.

2.4.4 Learned Constellation Shaping

To understand how PRNet achieves this implicit coding gain while simultaneously reducing PAPR, it is instructive to examine the raw output of the encoder. Unlike classical OFDM, which enforces rigidly discrete QPSK boundaries, the PRNet encoder learns to dynamically distort and disperse the constellation points.

Figure 2.6 visualizes the complex output symbols generated by the trained encoder. By mapping the discrete input data into this continuous, cloud-like spatial distribution, the network successfully trades off traditional Euclidean spacing to achieve optimal PAPR reduction and joint source-channel coding over the Rayleigh fading channel.

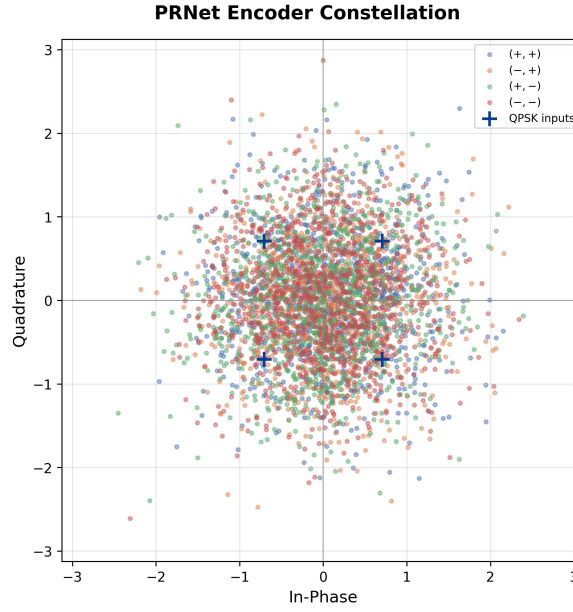


FIGURE 2.6: The learned constellation output of the PRNet encoder. The network disperses the traditional QPSK boundaries into a continuous distribution to minimize PAPR while preserving decodability.

2.4.5 Computational Complexity

At inference time, PRNet eliminates the need for iterative searching (like in PTS) and replaces it with a single neural network forward pass.

The original authors reported the computational complexity of this forward pass as $\mathcal{O}(L \cdot K \cdot M)$, where L is the number of fully connected processing layers (5), K is the hidden layer size (2048), and M is the modulation order (4 for QPSK) [2]. However, this formulation contains a mathematical error. In a fully connected network, passing data from one hidden layer to the next requires multiplying a vector of size K by a weight matrix of size $K \times K$. Because K is very large (2048), these hidden layer computations completely dominate the processing time. Therefore, the true theoretical complexity is $\mathcal{O}(L \cdot K^2)$, which is entirely independent of the modulation order M .

Despite this error in their theoretical formulation, the practical benefit remains clear: the computational cost of PRNet is fixed, meaning it does not scale exponentially with the search space like classical scrambling methods do.

TABLE 2.2: Execution time comparison for different PAPR reduction methods, as reported by the original authors [2].

Method	Execution time (μ s)
PTS	2,890
Clipping and filtering	1,415
PRNet (without parallelism)	2,304
PRNet (with GPU)	480

As shown in Table 2.2, applying GPU acceleration allows PRNet to run approximately 6 times faster than PTS and 3 times faster than clipping and filtering, all while delivering superior BER and PAPR performance [2].

2.5 Limitations

Despite its contributions, PRNet presents several limitations that define the research agenda for subsequent work:

- **Scalability Constraints:** Fully connected layers scale quadratically with the subcarrier count (N). While manageable for experimental $N = 64$ setups, scaling to commercial $N = 1024$ standards triggers a parameter explosion and prohibitive memory overhead, restricting deployment in large-scale systems.
- **Channel Mismatch Vulnerability:** The network severely overfits to the specific channel model used during training. Deploying PRNet in dynamic environments with unmodelled fading or imperfect channel state information degrades performance significantly.
- **Configuration Dependence:** The architecture relies on rigidly fixed input and output dimensions. Any real-time adjustment to the modulation order (M) or subcarrier count (N) breaks the model, requiring a full architectural redesign and retraining from scratch.
- **Lack of Interpretability:** The constellation shaping strategies are hidden within millions of "black-box" weights. This lack of transparency makes it mathematically impossible to provide the worst-case performance guarantees required by strict telecommunication standards.

APPENDIX A

Simulation Parameters

This appendix consolidates the simulation and system parameters used in the experiments presented in this thesis. Additional parameter tables will be added as further machine learning methods are investigated.

A.1 PRNet Parameters

The following table summarises the parameters adopted for the PRNet experiments, as reported by Kim et al. [\[2\]](#).

TABLE A.1: System and training parameters for PRNet [2].

Parameter	Value
<i>OFDM System</i>	
Number of subcarriers (N)	64
Modulation	QPSK
Oversampling factor (L)	4
Channel (training)	AWGN / Rayleigh flat fading
<i>Network Architecture</i>	
Hidden layers per DNN (encoder/decoder)	5
Neurons per hidden layer	2048
Hidden activation	ReLU
Output activation	Linear
Batch Normalization	Yes (after every FC layer)
<i>Training Configuration</i>	
Optimizer	Adam
Learning rate	10^{-4}
Batch size	400
Training steps	500,000
Optimal corruption level (η^*)	-15 dB
PAPR loss weight (λ)	0.01
<i>Reported Results</i>	
PAPR reduction at CCDF = 10^{-3}	$\approx 6-7$ dB
Inference speedup vs. PTS (GPU)	$\approx 6\times$

APPENDIX B

Glossary of Deep Learning Terms

This appendix provides concise definitions of deep learning concepts referenced throughout the document, intended as a quick reference for readers with a communications background.

Neuron

The basic computational unit of a neural network. It computes a weighted sum of its inputs, adds a bias, and applies an activation function: $y = f(\mathbf{w}^T \mathbf{x} + b)$.

Weight

A learnable parameter in a neural network that scales the connection between two neurons. The collection of all weights (and biases) constitutes the model's trainable parameters, denoted θ .

Bias

A learnable constant added to the weighted sum in a neuron before the activation function is applied: $z = \mathbf{W}\mathbf{x} + \mathbf{b}$. It allows the network to shift the activation function, giving it more flexibility.

Activation Function

A nonlinear function applied element-wise to the output of a neural network layer. Common choices include ReLU, sigmoid, tanh, softmax, and the linear (identity) function. Without nonlinear activations, a multi-layer network would collapse to a single linear transformation.

Fully Connected (FC) Layer

A layer where every neuron is connected to every neuron in the previous layer via a weight matrix $\mathbf{W}$. Also called a dense layer. The computational cost of an FC layer scales as $\mathcal{O}(n_{\text{in}} \times n_{\text{out}})$.

Forward Pass

The computation that propagates input data through the network, layer by layer, to produce an output. During inference (deployment), only the forward pass is needed—backpropagation is not required.

Inference

The process of using a trained model to make predictions on new, unseen data.

In the PAPR context, inference means performing one forward pass per OFDM symbol at real-time rates.

Loss Function

A scalar function that quantifies how far the network's output is from the desired output. Training minimises this function. Common choices include MSE for regression and cross-entropy for classification. In PAPR, custom losses combine PAPR and MSE terms.

Training

The iterative process of adjusting network parameters to minimise the loss function. Involves repeated forward passes, loss computation, backpropagation, and parameter updates over many epochs.

Gradient

The vector of partial derivatives of the loss function with respect to each trainable parameter. It indicates the direction and magnitude of the steepest increase in loss; the optimizer updates parameters in the opposite direction.

Backpropagation

The algorithm used to compute gradients of the loss function with respect to all network parameters by applying the chain rule of calculus, layer by layer, from the output back to the input. These gradients are then used by the optimizer to update the weights.

Optimizer

The algorithm that updates network parameters based on computed gradients. SGD (Stochastic Gradient Descent) is the simplest; Adam (Adaptive Moment Estimation) is the most commonly used in practice, combining momentum and per-parameter learning rate adaptation.

Learning Rate (η)

A hyperparameter that controls the step size of each parameter update. Too large a value causes training to diverge; too small a value makes convergence extremely slow.

Epoch

One complete pass through the entire training dataset. Training typically requires hundreds of epochs before the network converges to a good solution.

Batch

A subset of the training data processed together in one forward and backward pass. Using batches (rather than the entire dataset) enables efficient GPU utilisation and introduces beneficial stochastic noise into training.

Convergence

The point during training at which the loss function stabilises and further epochs yield negligible improvement. A model that fails to converge may be underfitting (too simple) or have a learning rate that is too high.

Hyperparameter

A configuration value set before training begins and not learned from data. Examples include learning rate, batch size, number of layers, number of epochs, and the loss balance factor α .

Overfitting

When a network learns to memorise the training data rather than generalise to unseen data. Symptoms include very low training loss but high validation/test loss. Mitigated by dropout, regularisation, and early stopping.

Regularisation

Any technique that prevents overfitting by constraining the model's complexity. Examples include dropout, weight decay (L_2 regularisation), and batch normalization.

Dropout

A regularisation technique that randomly sets a fraction of neuron outputs to zero during each training step. This prevents neurons from co-adapting and encourages the network to learn redundant representations, reducing overfitting.

Batch Normalization (BatchNorm)

A technique that normalises the inputs to each layer by subtracting the batch mean and dividing by the batch standard deviation. This stabilises training, allows higher learning rates, and acts as a mild regulariser.

Vanishing Gradients

A problem where gradients become extremely small as they propagate through many layers during backpropagation, causing early layers to learn very slowly or not at all. ReLU activations and residual connections help mitigate this.

Autoencoder

A neural network architecture consisting of an encoder that maps input data to a lower-dimensional (or transformed) representation, and a decoder that reconstructs the original data. In the PAPR context, the encoder produces a low-PAPR signal and the decoder recovers the original data symbols.

References

- [1] B. S. d. C. da Silva, V. D. Souto, R. D. Souza, and L. L. Mendes, “A survey of papr techniques based on machine learning,” *Sensors*, vol. 24, no. 6, p. 1918, 2024.
- [2] M. Kim, W. Lee, and D.-H. Cho, “A novel papr reduction scheme for ofdm system based on deep learning,” *IEEE Communications Letters*, vol. 22, no. 3, pp. 510–513, 2017.